`bispectrum`: Selective G-Bispectra Made Practical

Anonymous Author(s)
Affiliation
Address
email

Abstract

Many machine-learning tasks are invariant under group actions: image classification under planar rotation $\mathrm{SO}(2)$, signal classification under translation, and spherical-image classification under 3D rotations $\mathrm{SO}(3)$. The G-bispectrum is a principled *complete* invariant (it preserves all signal information up to the group action) but its deployment has been hampered by quadratic cost and a patchwork of group-specific implementations. We present `bispectrum`, an open-source, fully unit-tested PyTorch library that computes *selective* G-bispectral invariants for seven different group/domain pairs as drop-in differentiable `nn.Modules`. Selectivity reduces the invariant size from $O(|G|^2)$ for the full bispectrum to $O(|G|)$ while remaining complete on generic signals. On the theoretical side, we contribute a selective $\mathrm{SO}(3)$ invariant on the sphere S^2 that achieves $\Theta(L^2)$ coefficients at band-limit L, matching the information-theoretic orbit-space lower bound $(L{+}1)^2 - 3$ instead of the $O(L^3)$ of the full bispectrum. We provide proof-of-concept evidence that the construction retains enough information to recover signals up to rotation: gradient-descent reconstruction from the selective coefficients converges to rotated copies of the source signal across multiple test signals (Section 6.3). We benchmark the library on three datasets across three group/domain pairs — PCam under C_8, OrganMNIST3D under the octahedral group, and Spherical MNIST under $\mathrm{SO}(3)$ — comparing against norm pooling, gated pooling, Fourier-ELU pooling, max pooling, and (non-equivariant) augmented CNN baselines. The library delivers near-exact rotation invariance (test-set rotation-accuracy std $\sigma_{\mathrm{rot}} < 10^{-3}$) and a data-efficiency advantage over incomplete alternatives in the low-data, moderate-capacity regime; the selective forward pass runs in sub-millisecond time on GPU for finite groups. Code is available at `https://anonymous.4open.science/r/bispectrum/`.

1 Introduction

Many machine-learning tasks are invariant under the action of a symmetry group G: histopathology classification under planar rotations C_n, signal classification under translations, volumetric organ classification under cube rotations O, and spherical-image classification under 3D rotations $\mathrm{SO}(3)$. Equivariant neural networks [Cohen and Welling, 2016, Weiler and Cesa, 2019] enforce these symmetries by construction in their feature maps, but at the point where a prediction must be made any task that does not depend on the group action — classification, regression, detection — still has to map the equivariant features to an *invariant* representation.

The *G-bispectrum* is a complex-valued vector of triple-correlation coefficients of the group Fourier transform. It has three properties that matter for ML: (i) **completeness** — it preserves all signal information up to the group action (the map is one-to-one on group orbits); (ii) **inversion** — the signal can in principle be recovered from the invariant up to its orbit; and (iii) **selectivity** — a

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small $O(|G|)$ index subset of the $O(|G|^2)$ full bispectrum still suffices for completeness on generic signals [Mataigne et al., 2024].

Existing invariant pooling layers — norm pooling, gate-based pooling, max/sum aggregation [Weiler and Cesa, 2019] — are simple but *incomplete*: they collapse equivariant features to scalars and discard information that distinguishes signals in different orbits. The bispectrum offers a mathematically principled alternative [Giannakis, 1989, Kakarala, 2012] and has been used as a competitive pooling layer in equivariant networks [Sanborn et al., 2023, Mataigne et al., 2024].

Despite this, bispectral pooling has not been widely adopted, for both theoretical and computational reasons. Existing bispectral constructions are scattered across theory papers, domain-specific derivations, and small implementations, with no general-purpose PyTorch library covering the discrete and continuous symmetry groups relevant to modern geometric deep learning. Even for groups where selectivity [Mataigne et al., 2024] reduces the bispectrum from $O(|G|^2)$ to $O(|G|)$ scalars (and the per-pass cost from $O(|G|^3)$ to $O(|G|^2)$), a user still needs a numerically stable implementation, a consistent API, an inversion routine, tests, and benchmarked runtime. On the theoretical side, the selective construction of Mataigne et al. [2024] applies to finite groups; the continuous case of $\mathrm{SO}(3)$ acting on the sphere S^2 has remained open.

This paper presents `bispectrum`, a PyTorch [Paszke et al., 2019] library for computing bispectral invariants and, where available, their inverses. The library supports the C_n, D_n, $\mathbb{T}^d$, disk, S^2, and octahedral domains as differentiable `nn.Modules` intended to plug into standard ML workflows. Beyond the software contribution, the paper makes one technical contribution that arose during implementation: a selective $\mathrm{SO}(3)$ bispectrum on S^2 with $\Theta(L^2)$ coefficients (Section 6).

Contributions.

1. **Software contribution.** A PyTorch library implementing selective G-bispectra for *seven different group/domain pairs* (a *domain* is the underlying space on which the group acts: C_n on C_n, $\mathrm{SO}(2)$ on S^1, $\mathbb{T}^d$, D_n, $\mathrm{SO}(2)$ on the disk, $\mathrm{SO}(3)$ on S^2, and the octahedral group O) as drop-in differentiable `nn.Modules` with a consistent API, GPU-optimized forward passes ($<$1 ms for finite groups), precomputed CG buffers, and a fully unit-tested codebase (96.8% line coverage).
2. **Selective $\mathrm{SO}(3)$ bispectrum on S^2.** We construct an invariant on S^2 that uses $\Theta(L^2)$ scalar coefficients (matching the orbit-space dimension $(L+1)^2-3$ up to a constant factor) instead of the $O(L^3)$ of the full bispectrum, extending the BFS-style finite-group construction of Mataigne et al. [2024] to the continuous case $\mathrm{SO}(3)$ on S^2 (Section 6). We demonstrate empirically that the construction retains enough information to recover band-limited signals up to rotation: gradient-descent reconstruction from β_{sel} converges to rotated copies of the source signal across all test cases (Figure 5). A formal proof of completeness for the selective construction is left to future work.
3. **Empirical validation.** A benchmark of G-bispectral pooling across three ML tasks — PCam under C_8, OrganMNIST3D under the octahedral group O, and Spherical MNIST under $\mathrm{SO}(3)$ — against norm pooling, gated pooling, Fourier-ELU pooling, max pooling, and (non-equivariant) augmented-CNN baselines. Complete bispectral invariants provide a data-efficiency advantage in the low-data, moderate-capacity regime while converging to incomplete alternatives at high capacity.

Together, these contributions provide the first general-purpose, drop-in differentiable library for G-bispectral invariants and the first practical extension of the selective construction from finite groups to $\mathrm{SO}(3)$ on S^2.

2 Mathematical Background

Notation. A *group* G is a set with an associative product, identity, and inverse (e.g. rotations of the plane, translations of an image grid). G is *abelian* if elements commute (e.g. cyclic groups C_n, the torus $\mathbb{T}^d$) and *non-abelian* otherwise (e.g. $\mathrm{SO}(3)$, the dihedral groups D_n, the octahedral group O). A *representation* ρ assigns to each $g \in G$ a unitary matrix $\rho(g)$ such that $\rho(gh) = \rho(g)\rho(h)$; an *irreducible representation* (*irrep*) admits no nontrivial invariant subspace. The *dual* $\hat{G}$ denotes the

Table 1: Invariant nonlinearities for equivariant CNNs. "Exact"* means the invariant has no discretisation/aliasing error; "Complete" means it separates orbits; the coefficient column counts the size of the output for a signal with $|G|$ Fourier samples. "Genericity assumption"† means completeness holds outside a Lebesgue-measure-zero algebraic subvariety of the signal space. The selective bispectrum is the only method that is simultaneously exact, complete, and linear-cost for finite groups.

Nonlinearity	Exact?*	Complete?	Coefficients	Forward cost	Key limitation
Norm	Yes	No	$O(\|G\|)$	$O(\|G\|)$	Discards phase
Gate	Yes	No	$O(\|G\|)$	$O(\|G\|)$	Scalar bottleneck
Fourier pointwise	No	Yes‡	$O(\|G\|)$	$O(\|G\| \log \|G\|)$	Aliasing
Full bispectrum	Yes	Yes	$O(\|G\|^2)$	$O(\|G\|^3)$	Quadratic cost
Selective bispectrum	**Yes**	**Yes**	$\mathbf{O(\|G\|)}$	$\mathbf{O(\|G\|^2)}$	Genericity assumption†

‡Complete pre-aliasing; the pointwise nonlinearity in the forward pass introduces aliasing that may break completeness in practice.

set of equivalence classes of irreps; for finite G, $|\hat{G}| = O(|G|)$. A *group action* of G on a domain X is a map $G \times X \to X$, $(g, x) \mapsto g \cdot x$, compatible with the group product; it induces an action on functions $f : X \to \mathbb{R}$ by $(f \circ g)(x) := f(g \cdot x)$. Two signals are in the same *orbit* if one is the image of the other under some $g \in G$; an *invariant* of f is a function constant on each orbit. For compact G there is a unique normalised translation-invariant measure $d\mu$ called the *Haar measure*; finite G is the special case where $d\mu$ is the counting measure. "*Lebesgue-almost-every f*" means "every f outside a measure-zero algebraic subvariety of the signal space." The groups and domains used in this paper are: the cyclic group C_n (discrete planar rotations), the dihedral group D_n (rotations and reflections of a regular n-gon), the torus $\mathbb{T}^d = (S^1)^d$ (multi-channel periodic signals), $\mathrm{SO}(2)$ acting on the disk (continuous planar rotations of a 2D image), $\mathrm{SO}(3)$ acting on the sphere S^2 (continuous 3D rotations of a spherical signal), and the chiral octahedral group O (24 proper rotations of the cube, acting on 3D voxel grids).

2.1 Equivariant features and invariant maps

A neural network layer Φ is *G-equivariant* if its output transforms predictably under the group action: $[\pi_{\mathrm{out}}(g) \circ \Phi](f) = \Phi(\pi_{\mathrm{in}}(g)f)$ for all $g \in G$ [Cohen and Welling, 2016, Weiler and Cesa, 2019]. Here f is an input signal (a function on the domain X), $\pi_{\mathrm{in}}, \pi_{\mathrm{out}}$ are the (representations of the) G-actions on the input and output feature spaces of Φ, and the composition $\pi_{\mathrm{out}}(g) \circ \Phi$ is the action of g on the output side. Stacking equivariant layers preserves equivariance, but any task that requires a prediction independent of the group action—classification, regression, detection—must eventually map equivariant features to *invariant* ones (constant on each orbit).

Table 1 summarises the trade-offs between exactness, completeness, and computational cost across existing invariant pooling layers. Norm pooling and gated nonlinearities are exact invariants but *incomplete*: they discard the phase of Fourier blocks, collapsing distinct orbits. Fourier-space pointwise nonlinearities preserve completeness but sacrifice exactness through aliasing [Weiler and Cesa, 2019]. The full bispectrum is exact and complete but costs $O(|G|^2)$ coefficients. The *selective* bispectrum (introduced below) preserves both exactness and completeness while using only $O(|G|)$ coefficients on generic signals.

2.2 The G-bispectrum and selective bispectra

Let G be a compact group with dual $\hat{G}$ and irreps $\{\rho_i\}_{i \in \hat{G}}$ of dimensions d_i. The *group Fourier transform* [Diaconis and Rockmore, 1990] of a signal $f : G \to \mathbb{C}$ is

$$\hat{f}(\rho_i) = \int_G f(g)\, \rho_i(g)\, d\mu(g) \quad \in \mathbb{C}^{d_i \times d_i}, \tag{1}$$

which specialises to $\sum_{g \in G} f(g)\, \rho_i(g)$ when G is finite (counting measure). The Clebsch–Gordan (CG) decomposition $\rho_i \otimes \rho_j \cong \bigoplus_{k \in \hat{G}} m_{ij}^k \rho_k$ expresses the tensor product of two irreps as a direct sum of irreps (with integer multiplicities $m_{ij}^k \geq 0$); the change-of-basis is realised by the *CG isometry* $C_{ij}^k : V_{\rho_k} \to V_{\rho_i} \otimes V_{\rho_j}$, projecting the tensor product onto its ρ_k-isotypic component. For each

admissible triple (i, j, k) with $m_{ij}^k > 0$, the *bispectrum* is the scalar

$$\beta_{i,j}^k(f) := \left\langle (C_{ij}^k)^\dagger \left(\hat{f}(\rho_i) \otimes \hat{f}(\rho_j)\right),\, \hat{f}(\rho_k)\right\rangle. \tag{2}$$

Each $\beta_{i,j}^k$ is invariant under the action $f \mapsto f \circ g$ and, taken over all admissible triples, the bispectrum $\{\beta_{i,j}^k\}_{(i,j,k)}$ is a *complete* invariant: it separates orbits for generic signals [Kakarala, 2012]. For finite G, $|\hat{G}| = O(|G|)$, so the full bispectrum has $O(|G|^2)$ scalar coefficients (and is naturally indexed by $(i, j) \in \hat{G}^2$ with k ranging over the CG-allowed irreps). Equation (2) is stated for G acting on itself; for homogeneous spaces of the form G/H (e.g. $\mathrm{SO}(3)$ on S^2, or $\mathrm{SO}(2)$ on the disk), the same construction applies after expanding f in a basis adapted to G/H (spherical harmonics for S^2, Bessel modes on the disk). The explicit specialisation for S^2 is in Definition 4.

Scalar vs. matrix (vector) bispectrum. Throughout this paper the term *scalar bispectrum* refers to the single complex number $\beta_{i,j}^k(f)$ obtained from the inner product in Eq. (2)—i.e., the CG-contracted scalar associated with each admissible triple. In the literature one also encounters the *matrix* or *vector* bispectrum, which keeps the un-contracted matrix $(C_{ij}^k)^\dagger(\hat{f}(\rho_i) \otimes \hat{f}(\rho_j))$ before pairing it with $\hat{f}(\rho_k)$. The two are related by a single trace/inner-product step; the scalar form is what the library outputs and what we analyse for $\mathrm{SO}(3)$ on S^2.

Selective bispectra. Mataigne et al. [2024] show that for finite groups, a breadth-first search (BFS) over the Kronecker product table of $\hat{G}$ selects $O(|G|)$ bispectral coefficients $\beta_{i,j}^k$ that remain complete; Myers and Miolane [2025] extend this to the disk. This reduction from quadratic to linear is the key to practical deployment: for example, the bispectrum of the octahedral group drops from 576 to 172 coefficients.

3 Related Work

Equivariant network libraries. `escnn` [Cesa et al., 2022] and `e3nn` [Geiger and Smidt, 2022] provide general-purpose equivariant convolution layers for 2D and 3D groups respectively, and `lie-nn` [Batatia et al., 2023] provides general Lie-group representation primitives (irreducible representations, Clebsch–Gordan decompositions, tensor products) on top of which custom equivariant layers can be built. These libraries implement equivariant *feature maps* but leave the choice of invariant pooling to the user (typically norm or gate). Our library is complementary: it provides the invariant map as a drop-in `nn.Module` that plugs into architectures built with any of these frameworks.

Bispectral approaches in neural networks. Sanborn et al. [2023] introduced bispectral neural networks (BNNs), using the full G-bispectrum as an invariant nonlinearity for C_n and D_n. Oreiller et al. [2022] applied bispectral CNNs to medical imaging with a fixed set of hand-selected triples. Both use the full (quadratic-cost) bispectrum and provide implementations tied to specific architectures. Our library generalizes to seven group/domain pairs, defaults to the $O(|G|)$ selective reduction [Mataigne et al., 2024], and exposes a framework-agnostic API. Chevalley et al. [2024] explore bispectral signatures for data characterisation, using bispectra as descriptors rather than learnable pooling primitives. Outside the equivariant-network setting, `PyBispectra` [Binns et al., 2025] computes bispectrum and bicoherence statistics of one-dimensional time-series signals for electrophysiology applications such as phase-amplitude coupling and time-delay estimation; the underlying 1D bispectrum is translation-invariant by construction, but the toolbox is not designed as a neural-network pooling layer.

Bispectra on homogeneous and non-homogeneous spaces. The selective G-bispectrum construction [Mataigne et al., 2024] reduces the cost of the G-bispectrum from $O(|G|^2)$ to $O(|G|)$ scalars on finite groups. Myers and Miolane [2025] extend the selective construction to the disk, where $\mathrm{SO}(2)$ acts on a non-homogeneous space. Kakarala [2012] treats the $\mathrm{SO}(3)$-on-$\mathrm{SO}(3)$ bispectrum and the closely related sphere case via Wigner-D irreps. The matrix (vector) bispectrum on S^2 is the homogeneous-space form on which our scalar contractions and CG power augmentation are built.

Table 2: Library profiling summary. Timings are median wall-clock on a single NVIDIA H100 80 GB GPU. "Coefs" = number of scalar coefficients (Sel. = selective, Full = full bispectrum); "Fwd" = forward-pass time in milliseconds; "Inv" = inversion time in milliseconds. "–" = not implemented. For continuous groups, $|G| = \infty$; coefficient counts refer to the band-limited representation at the tested resolution. *`SO3onS2` benchmarked at band-limit $\ell_{\max}$=16 (we use $\ell_{\max}$ and L interchangeably to denote the band-limit; see Section 6).

Module	G	$\|G\|$	Coefs		Fwd (ms)		Inv (ms)
			Sel.	Full	Sel.	Full	Sel.
`CnonCn`	C_{128}	128	128	8256	0.15	9.17	8.56
`TorusOnTorus`	$C_{32} \times C_{32}$	1024	1024	524800	0.08	0.71	68.02
`DnonDn`	D_{32}	64	245	–	2.52	–	42.33
`SO2onDisk`	$\mathrm{SO}(2)$	∞	411	–	1.94	–	17.69
`SO3onS2`*	$\mathrm{SO}(3)$	∞	430	–	26.54	–	–
`OctaonOcta`	O	24	172	–	0.70	–	–

CG product nonlinearities and tensor products. Tensor field networks [Thomas et al., 2018] and MACE [Batatia et al., 2022] use iterated Clebsch–Gordan products to build equivariant interactions. These products are exact and complete in principle, but each layer incurs $O(L^3)$ cost per CG contraction. Xie et al. [2025] analyse the *price of freedom* in tensor product architectures, showing that maintaining full expressivity through CG products becomes expensive at high angular momenta. The selective bispectrum is empirically complete (proof of concept in Figure 5; finite-group case proved in Mataigne et al. [2024]) in a single $O(L^2)$ pass for $\mathrm{SO}(3)$ or $O(|G|)$ pass for finite groups, avoiding iterated contractions.

Invariant theory and scattering. The wavelet scattering transform [Mallat, 2012] provides translation-invariant representations via iterated modulus operations; Esteves et al. [2018] extend this to the sphere. Unlike the G-bispectrum, scattering representations are stable but generally incomplete—they discard inter-scale phase information. The G-bispectrum's completeness guarantee [Giannakis, 1989, Kakarala, 2012] is its distinguishing theoretical advantage.

4 The `bispectrum` Library

4.1 Design and implementation

Every group/domain pair is exposed as a `torch.nn.Module` named `{Group}on{Domain}` (e.g. `SO3onS2`). Each module exposes a uniform API: `forward(f)` returns the selective bispectral invariants of f; `fourier(f)` returns the underlying group Fourier coefficients; and, where available, `invert(beta)` returns the reconstructed signal up to group-action indeterminacy. Modules accept raw real-valued inputs, handle the Fourier transform internally, and default to $O(|G|)$ selective coefficients (`selective=False` gives the full $O(|G|^2)$ set). Clebsch–Gordan (CG) matrices, discrete Fourier transform (DFT) kernels, and Bessel roots (used for the radial discretisation of the disk in `SO2onDisk`) are precomputed at construction and stored as non-learnable buffers; this lets `.to(device)` migrate the module transparently and ensures that forward passes incur zero recomputation. The only dependencies are PyTorch [Paszke et al., 2019], NumPy [Harris et al., 2020], and `torch_harmonics` [Bonev et al., 2023] (for `SO3onS2`). For $\mathrm{SO}(3)$ on S^2, reduced CG matrices are built in parallel and cached to disk; all forward methods are batch-native, using `torch.einsum`/`torch.bmm` for non-abelian modules and `torch.fft` for abelian ones. The test suite covers 96.8% of lines across all modules. A minimal usage example is in Appendix F.

4.2 Profiling

We profile the library on a single NVIDIA H100 80 GB GPU (PyTorch 2.10, batch = 16 for the forward pass and 4 for inversion, median wall-clock via `torch.utils.benchmark`); results are in Table 2 and Figure 1.

The selective forward pass is sub-millisecond for finite abelian modules (C_n: 0.15 ms, $\mathbb{T}^2$: 0.08 ms), under 2 ms for $\mathrm{SO}(2)$/disk (1.94 ms), and under 1 ms for the octahedral group (0.70 ms), making it

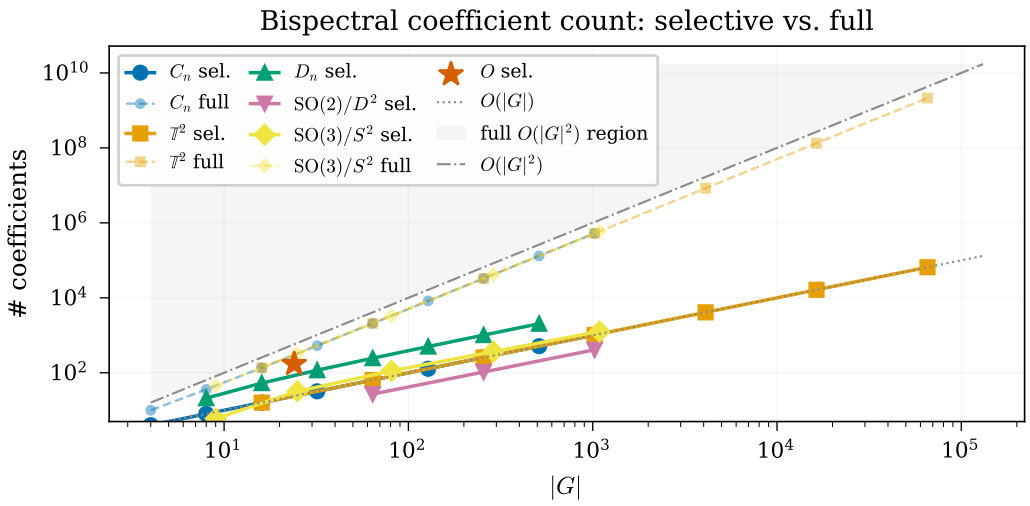

Figure 1: Bispectral coefficient count vs. group order $|G|$. Solid lines: selective ($O(|G|)$); dashed: full ($O(|G|^2)$). At $|G|$=1,024, selectivity yields a 512× reduction. Continuous groups (SO(2), SO(3)) are plotted at their band-limited signal dimension. The full-bispectrum curves for D_n and the disk are omitted because the corresponding backends are not yet implemented in the library.

negligible relative to backbone computation. SO(3) on S^2 at $\ell_{\max}$=16 takes 26.5 ms, dominated by the spherical harmonic transform (via `torch_harmonics` [Bonev et al., 2023]). Selective coefficient counts confirm the $O(|G|)$ scaling: C_{128} uses 128 coefficients vs. 8,256 full (64× reduction); $\mathbb{T}^2$ at $|G|$=1,024 uses 1,024 vs. 524,800 full (512×). Inversion is fast for abelian groups (9–68 ms). GPU throughput scales linearly with batch size, exceeding 10^7 samples/s for C_n and $\mathbb{T}^2$ (Appendix B).

5 Experiments

We validate the library on three tasks spanning discrete and continuous groups: 2D histopathology classification under C_8 symmetry (PCam, Section 5.1), 3D organ classification under octahedral symmetry (OrganMNIST3D, Section 5.2), and spherical digit classification under full SO(3) (Spherical MNIST, Section 5.3).

5.1 2D histopathology classification under cyclic symmetry

We evaluate five invariant pooling strategies on PatchCamelyon [Veeling et al., 2018] (PCam), 327K histopathology patches with binary metastasis labels. All equivariant models share a C_8-equivariant DenseNet [Huang et al., 2017] backbone following Veeling et al. [2018]; they differ only in the terminal invariant map: norm pooling, gated pooling, Fourier-ELU, and selective bispectrum via `CnonCn`(n=8). A non-equivariant standard CNN trained with data augmentation drawn uniformly from the symmetry group C_8 ("*augmented CNN*") serves as the non-equivariant baseline. We sweep growth rates to vary capacity (30K–1.6M params) and train on 10% of the PCam training set (an image-level subsample, not a group-orbit subsample, drawn before any group augmentation) with AdamW [Loshchilov and Hutter, 2019]. Full model descriptions are in Appendix C.

Results. Table 3 and Figure 2 present the results at matched ∼100K parameters.

At matched capacity (∼100K params, 10% data), all four equivariant pooling methods substantially outperform the standard augmented CNN (AUC 0.896). Fourier-ELU leads slightly (0.945), followed by NormReLU (0.942), bispectrum and gated (both 0.941). At 1% data, the bispectrum leads all methods (0.912), outperforming NormReLU (0.873) and Fourier-ELU (0.876). This data-efficiency advantage narrows as the training set grows, consistent with complete invariants providing a stronger inductive bias when data is scarce.

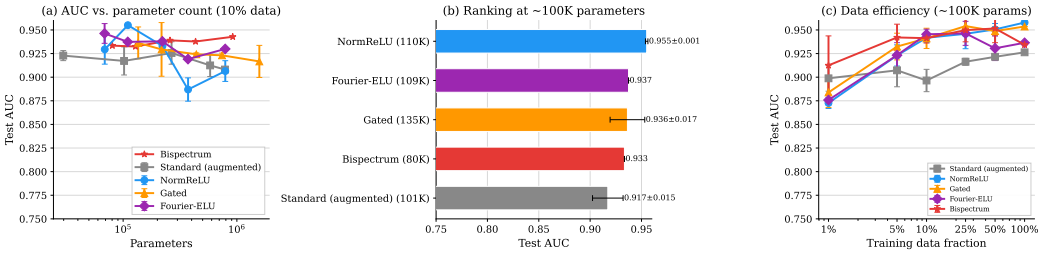

Figure 2: PCam Pareto analysis (`CnonCn` selective bispectrum, single seed, 10% training data). (a) Test AUC vs. parameter count across all five pooling variants. (b) Ranking at the ∼100K parameter budget. (c) Data efficiency at matched ∼100K parameters. Multi-seed statistics are in Table 3.

Table 3: PCam results at matched ∼100K parameters (10% training data, mean ± std over 3 seeds). All equivariant models share a C_8-equivariant DenseNet backbone.

Model	Params	Test AUC	Test ACC
Standard (augmented)	102K	.896 ± .010	.826 ± .012
Gated	136K	.941 ± .009	.870 ± .004
Selective Bispectrum (`CnonCn`)	128K	.941 ± .004	**.872 ± .006**
Fourier-ELU	110K	**.945 ± .004**	.855 ± .006
NormReLU	110K	.942 ± .004	.858 ± .007

5.2 3D organ classification under octahedral symmetry

To evaluate the `OctaonOcta` module for the chiral octahedral group O ($|O| = 24$), we design a controlled 3D experiment on OrganMNIST3D [Yang et al., 2023] (1,742 abdominal CT volumes, 28^3 voxels, 11 classes). All equivariant variants share an O-equivariant 3D ResNet backbone with exact voxel-permutation kernels. The four models compared are: a translation-equivariant standard 3D CNN trained with octahedral data augmentation (the "*augmented CNN*" baseline, 16K params); the same O-equivariant ResNet with terminal norm pool (375K params); with terminal max pool (374K params); and with terminal selective bispectrum via `OctaonOcta` (463K, 172 invariant scalars). Models are trained with AdamW [Loshchilov and Hutter, 2019] for 100 epochs with three seeds; rotation robustness is measured over all 24 octahedral rotations of the test set. Published baselines and wider-channel experiments are in Appendix D.

Results. Table 4 and Figure 3 report the key findings.

Rotation invariance. All equivariant models achieve $\sigma_{\text{rot}} \leq 0.0005$, while the augmented CNN drops ∼2.5 pp (percentage points) under rotation.

Completeness vs. lossy pooling. The bispectrum (72.6%) matches max pool (73.0%) and substantially outperforms norm pool (56.8%) and Regular SE(3) convolutions [Kuipers and Bekkers, 2023] (69.8%). At reduced data (Figure 3), the bispectrum's advantage widens: at 5% data it reaches 33.2% vs. 18.9% for max pool. At higher capacity (6M params, Appendix D), max pool overtakes the bispectrum, suggesting the completeness advantage is specific to the data-limited regime.

5.3 Spherical MNIST: SO(3) invariance on the sphere

The preceding experiments validate bispectral pooling for finite groups (C_8 and the octahedral group O). To evaluate the library's continuous group support, we test `SO3onS2`—the $\mathrm{SO}(3)$ bispectrum on the 2-sphere—on Spherical MNIST [Cohen et al., 2018], the standard benchmark for rotation invariance on S^2.

Setup. MNIST digits are projected onto the sphere via stereographic projection and sampled on a 64×128 equiangular grid. We compute the selective bispectrum at $\ell_{\max} = 15$, producing 384 com-

Table 4: OrganMNIST3D results (mean $\pm$ std over 3 seeds). Rot ACC and σ_{rot}: mean and std of accuracy across 24 octahedral test-set rotations.

Model	Params	Test ACC	Test AUC	Rot ACC	σ_{rot}
Standard 3D CNN	16K	$.601 \pm .017$	$.951 \pm .002$	.598	.0115
O-Equiv + Norm Pool	375K	$.568 \pm .099$	$.940 \pm .021$	.568	.0000
O-Equiv + Max Pool	374K	$.730 \pm .033$	$.972 \pm .005$	.730	.0000
O-Equiv + `OctaonOcta` Bispectrum	463K	$.726 \pm .027$	$.972 \pm .004$	.726	.0005†

†Nonzero σ_{rot} due to floating-point precision in CG contractions, not a theoretical invariance failure.

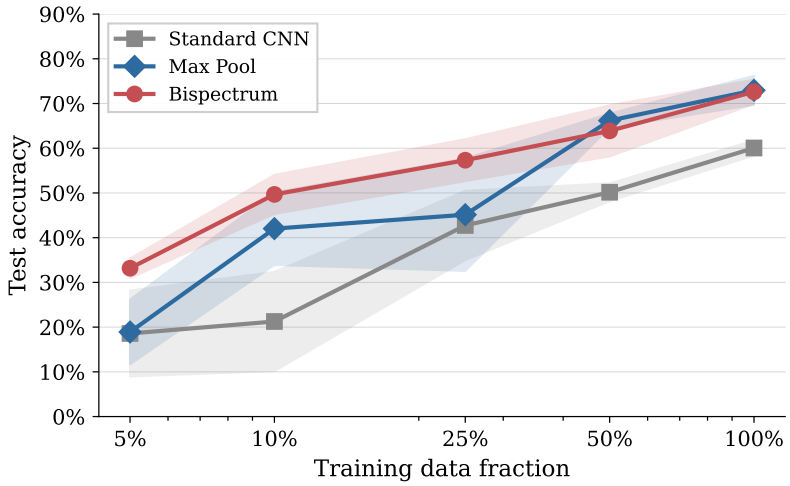

Figure 3: Test accuracy vs. training data fraction on OrganMNIST3D (`OctaonOcta` bispectrum, mean over 3 seeds; shaded bands: ± 1 std). The complete invariant map provides a consistent advantage in the low-data regime.

plex invariant coefficients. To form the network input we apply a sign-preserving $\operatorname{sign}(x)\log(1+|x|)$ transform separately to the real and imaginary parts of each coefficient, yielding 768 real features. By Lemma 14, exactly one of the two log1p features per coefficient is structurally zero on real signals (real for even parity, imaginary for odd parity); we keep both channels because the transform is sign-preserving and per-channel injective. The features are fed to a 3-layer MLP (232K parameters total). Three baselines provide controls:

- **Power Spectrum + MLP** (11K params): computes only the rotation-invariant power spectrum $\|f_\ell\|^2$ per degree ℓ, discarding all phase coupling between harmonics. Invariant but *incomplete*.
- **Power Spectrum + MLP (matched)** (166K params): same incomplete invariant features but with a larger MLP (166K), isolating the effect of feature completeness from model capacity.
- **Standard CNN** (185K params): a conventional CNN operating directly on the equirectangular image. Not rotation-invariant.

Following Cohen et al. [2018], we evaluate under three protocols: train on non-rotated / test on non-rotated (NR/NR), train on rotated / test on rotated (R/R), and the most challenging train on non-rotated / test on rotated (NR/R). Each configuration is run with three seeds. Rotation robustness is assessed by applying 10 random $\mathrm{SO}(3)$ rotations to the entire test set and measuring accuracy variance.

Results. Table 5 and Figure 4 report the results.

Table 5: Spherical MNIST results (mean $\pm$ std over 3 seeds). Reference numbers are from Cohen et al. [2018]. σ_{rot}: standard deviation of accuracy under 10 random $\mathrm{SO}(3)$ rotations (NR-trained models).

Model	Params	NR/NR ↑	R/R ↑	NR/R ↑	σ_{rot} ↓
Cohen planar CNN	—	0.98	0.23	0.11	—
Cohen spherical CNN	—	0.96	0.95	0.94	—
Standard CNN	185K	0.991±0.000	0.781±0.003	0.138±0.003	0.044
Power Spec. + MLP	11K	0.780±0.002	0.780±0.002	0.777±0.001	3×10^{-4}
Power Spec. + MLP (matched)	166K	0.794±0.011	0.794±0.011	0.792±0.010	3×10^{-4}
`SO3onS2` **Bispectrum + MLP (ours)**	232K	**0.950±0.000**	**0.949±0.002**	**0.951±0.001**	$\mathbf{1\times10^{-4}}$

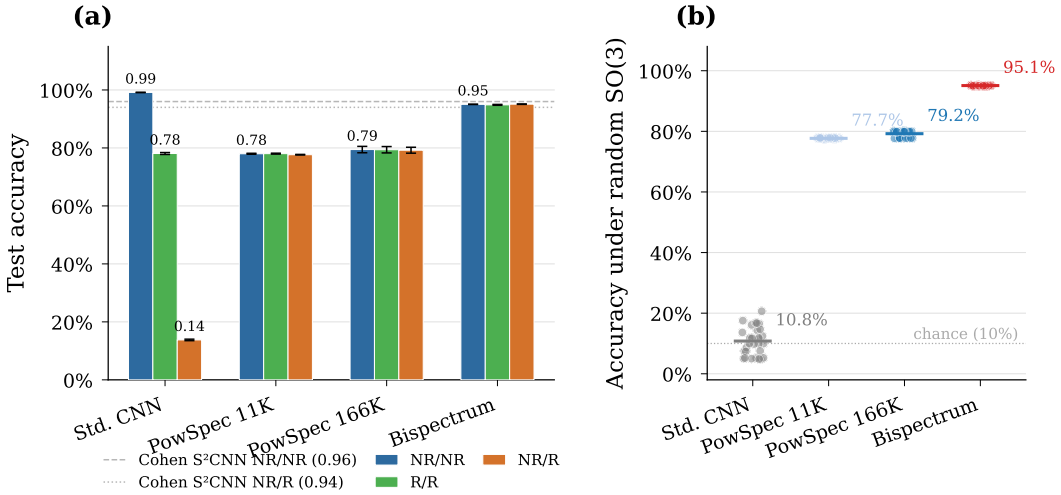

Figure 4: Spherical MNIST analysis (`SO3onS2` selective bispectrum). **(a)** Cohen et al. protocol: NR/NR, R/R, and NR/R accuracy. Dashed lines show Cohen's spherical CNN reference. **(b)** Accuracy under 10 random $\mathrm{SO}(3)$ rotations (NR-trained models, 30 points per model = 10 rotations $\times$ 3 seeds). Scaling the power spectrum MLP from 11K to 166K parameters yields only +1.4pp; the `SO3onS2` bispectrum at 232K reaches 95.0%.

Numerically exact $\mathrm{SO}(3)$ invariance. The bispectrum achieves near-identical accuracy across all three protocols ($\sigma_{\text{rot}} \sim 10^{-4}$), confirming true $\mathrm{SO}(3)$ invariance. The standard CNN collapses from 99.1% (NR/NR) to 13.8% (NR/R), showing that augmentation alone does not confer invariance.

Completeness gap. The power spectrum is equally invariant but reaches only 78%—a 17pp deficit that persists even after scaling the MLP to 166K parameters (+1.4pp), confirming incompleteness rather than capacity as the bottleneck.

Comparison with Cohen et al. Cohen's spherical CNN [Cohen et al., 2018] reaches 96%/95%/94% with learned equivariant convolutions. Our single-pass bispectrum (95.0%/94.9%/95.1%) is 1–2pp lower but has a tighter invariance gap (0.1pp vs. 2.0pp).

6 Selective $\mathrm{SO}(3)$ bispectrum on S^2

6.1 Setup

A real signal $f : S^2 \to \mathbb{R}$ band-limited at degree L expands in spherical harmonics as $f(\theta, \varphi) = \sum_{\ell=0}^{L} \sum_{m=-\ell}^{\ell} a_\ell^m Y_\ell^m(\theta, \varphi)$ with the reality constraint $a_\ell^{-m} = (-1)^m \overline{a_\ell^m}$, so it is parameterised by $(L+1)^2$ real scalars (Definition 4, eq. (5)). On $S^2 = \mathrm{SO}(3)/\mathrm{SO}(2)_z$, the relevant irreducible representations of $\mathrm{SO}(3)$ are $\{\rho_\ell\}_{\ell\geq 0}$ with V_{ρ_ℓ} the $(2\ell+1)$-dimensional space of degree-ℓ spherical

harmonics and Wigner-D action $\rho_\ell(g) = D^\ell(g)$. In the irrep notation of Mataigne et al. [2024], the Fourier coefficient at ρ_ℓ is $\mathcal{F}(f)_{\rho_\ell} = \mathbf{F}_\ell \in \mathbb{C}^{2\ell+1}$; we write $\mathbf{F}_\ell$ throughout. The scalar ρ_ℓ-isotypic component of the matrix-valued bispectrum $\beta_{\rho_{\ell_1},\rho_{\ell_2}}(f)$ is the familiar

$$\beta_{\ell_1,\ell_2,\ell} = \sum_{\substack{m_1,m_2,m \\ m_1+m_2=m}} C^{\ell,\,m}_{\ell_1,\,m_1;\,\ell_2,\,m_2}\, a^{m_1}_{\ell_1}\, a^{m_2}_{\ell_2}\, \overline{a^{m}_{\ell}}, \tag{3}$$

the specialisation of eq. (2) to $\mathrm{SO}(3)$-on-S^2. The full bispectrum has $\Theta(L^3)$ such scalars; the orbit space $\mathbb{R}^{(L+1)^2}/\mathrm{SO}(3)$ has dimension $(L+1)^2 - 3$, so any complete invariant requires $\Omega(L^2)$ independent components (Proposition 8). The Mataigne chain $\beta_{1,\ell,\ell+1}$ for $\ell = 0, \ldots, L-1$ provides only $\sim 3L$ scalars and cannot achieve this bound.

6.2 Selective construction and algorithm

Let $L_{\mathrm{seed}} := \min(L, 4)$. The *seed index set* collects all admissible bispectral triples up to L_{seed},

$$\mathcal{S}_{\mathrm{seed}} := \{(\ell_1, \ell_2, \ell) : 0 \le \ell_1 \le \ell_2 \le L_{\mathrm{seed}},\ |\ell_1 - \ell_2| \le \ell \le \min(\ell_1 + \ell_2, L_{\mathrm{seed}})\},$$

of constant size ≤ 50 once $L \ge 4$, matching the implementation seed (Algorithm 2). For each high-degree target $\ell \ge 5$, we use a size-$(2\ell+1)$ *bootstrap block* $\mathcal{T}_\ell$, which for $\ell \ge 8$ is the closed-form family of eq. (8) (Definition 11) and for $5 \le \ell \le 7$ is the explicit small block $\mathcal{T}^{\mathrm{small}}_\ell$ tabulated in Algorithm 2. Both versions consist of entries linear in $\mathbf{F}_\ell$ with coefficients depending only on already-recovered lower degrees. The *selective* $\mathrm{SO}(3)$ *bispectrum on* S^2 is then the concatenation

$$\beta_{\mathrm{sel}}(f) := \left\{\beta_{\ell_1,\ell_2,\ell}\right\}_{(\ell_1,\ell_2,\ell)\in\mathcal{S}_\beta}, \qquad \mathcal{S}_\beta := \mathcal{S}_{\mathrm{seed}} \cup \bigcup_{\ell=5}^{L} \mathcal{T}_\ell. \tag{4}$$

By Lemma 14, each scalar entry contributes exactly one real degree of freedom on the real-signal slice (real for even parity, purely imaginary for odd parity); we record that single real number per entry. Algorithm 1 writes this as a Mataigne-style BFS over selected triples; the implementation-grade index set (Algorithm 2 in Appendix A) augments $\mathcal{S}_\beta$ with mandatory even self-couplings and optional CG-power scalars for numerical robustness, and is a strict superset of β_{sel}.

Algorithm 1 Selective $\mathrm{SO}(3)$ bispectrum on S^2 (BFS form, mirroring Mataigne et al. [2024], Alg. 1).

```
Input: band-limit L, seed cutoff L_seed = min(L, 4)
L_β ← all admissible triples (ℓ1, ℓ2, ℓ) with ℓ ≤ L_seed {seed block S_seed}
for ℓ = L_seed+1, ..., L do
    append the bootstrap block T_ℓ to L_β {2ℓ+1 entries linear in F_ℓ; T^small_ℓ for ℓ ≤ 7, eq. (8) for ℓ ≥ 8}
end for
return β_sel(f) = {β_{ℓ1,ℓ2,ℓ}(f) : (ℓ1, ℓ2, ℓ) ∈ L_β}
```

The output size is $|\beta_{\mathrm{sel}}| = |\mathcal{S}_{\mathrm{seed}}| + \sum_{\ell=5}^{L}(2\ell+1) = \Theta(L^2)$, matching the orbit-space lower bound up to a constant (Proposition 13).

Remark 1 (CG power augmentation in the implementation). The library appends the CG power spectrum $P_{\ell_1,\ell_2,\ell} := \|(\mathbf{F}_{\ell_1} \otimes \mathbf{F}_{\ell_2})|_\ell\|^2$ (Definition 5) at $O(L^2)$ extra cost. The pure linear bootstrap submatrix $A_\ell \in \mathbb{R}^{(2\ell+1)\times(2\ell+1)}$ assembled from $\mathcal{T}_\ell$ alone is rank-deficient on the real-signal slice (see Remark 20); appending the CG power scalars (and the mandatory even self-couplings of Algorithm 2) restores per-degree numerical robustness, and is what the implementation does unconditionally.

Motivation. Three structural facts justify this construction. (i) The output size $|\beta_{\mathrm{sel}}| = \Theta(L^2)$ matches the orbit-space dimension $(L+1)^2 - 3$ up to a constant factor (Proposition 8); no G-invariant of strictly lower complexity can be complete. (ii) The BFS-style construction directly mirrors the selective bispectrum of Mataigne et al. [2024] for finite groups, where the same $2\ell+1$ entries-per-shell pattern is provably complete; we extend their template from finite G to the continuous case $\mathrm{SO}(3)$ on S^2. (iii) The matrix bispectrum on $L^1(\mathrm{SO}(3))$ is known to be injective on dense orbits [Kakarala,

Table 6: Theorem–implementation consistency at $L = 15$ and $L = 16$. Formal $|\beta_{\text{sel}}|$ counts one real component per scalar entry (Lemma 14); the implementation `output_size` from `_build_selective_index_map` $\cup$ `_build_cg_power_index_map` is a strict superset, adding mandatory even self-couplings and CG-power scalars for numerical robustness; the Spherical MNIST feature vector applies a sign-preserving $\log 1p$ separately to the real and imaginary parts of each implementation scalar (Section 6).

L	Formal $\lvert\beta_{\text{sel}}\rvert$	Impl. `output_size`	SMNIST features
15	273	384	768
16	306	430	860

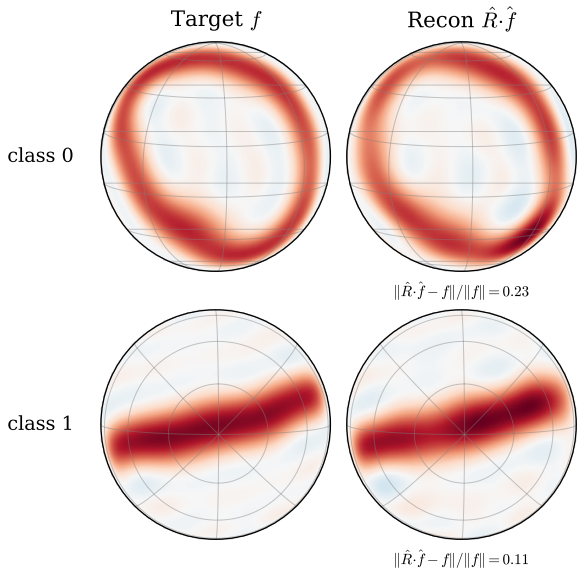

Figure 5: $\mathrm{SO}(3)$ bispectrum reconstruction (proof of concept). For each of two source signals (rows: Spherical MNIST class 0 and class 1, $L = 15$), the column *Target* f shows the original band-limited signal and the column *Recon* $\hat{R} \cdot \hat{f}$ shows the gradient-descent reconstruction $\hat{f}$ (recovered from $\beta_{\text{sel}}(f)$ only) after alignment to f by the optimal $\hat{R} \in \mathrm{SO}(3)$. The relative image residual $\|\hat{R} \cdot \hat{f} - f\|/\|f\|$ is shown beneath each row.

1992, 2012], suggesting the scalar CG contraction $\beta_{\ell_1,\ell_2,\ell}$ inherits enough information to separate orbits in the band-limited setting, at least generically. We do not give a formal proof of completeness here; instead we provide direct empirical evidence below, and leave a closed-form proof for future work.

6.3 Empirical proof of concept: bispectrum reconstruction

To test whether β_{sel} retains enough information to recover signals up to rotation, we run a direct inversion experiment. We take two band-limited signals $f : S^2 \to \mathbb{R}$ at $L = 15$ (one from each of the first two Spherical MNIST classes), compute $\beta_{\text{sel}}(f)$, and reconstruct f by gradient descent on the spherical-harmonic coefficients $\hat{a}_\ell^m$, minimising $\|\beta_{\text{sel}}(\hat{f}) - \beta_{\text{sel}}(f)\|^2$ from a random initialisation. After convergence, $\hat{f}$ is aligned to f by the optimal $\hat{R} \in \mathrm{SO}(3)$ (degree-1 alignment plus Wigner-D refinement).

Takeaways. On both signals the bispectrum residual reaches $\|\Delta\beta\|/\|\beta\| \sim 10^{-3}$ at convergence, and the aligned image residual $\|\hat{R} \cdot \hat{f} - f\|/\|f\|$ is 0.23 (class 0) and 0.11 (class 1); the recovered $\hat{f}$ visually matches f up to rotation in both cases (Figure 5). Residual image error is dominated by gradient-descent local optima rather than by missing invariant information: each reconstruction is initialised from a single random point and run with no restarts. We read this as proof-of-concept

evidence that the selective coefficients carry enough information to recover band-limited signals up to $\mathrm{SO}(3)$; a formal proof of completeness for β_{sel} remains an open question.

7 Discussion and Conclusion

We have presented `bispectrum`, a PyTorch library that makes selective G-bispectral invariants practical for machine learning. The library covers seven group/domain pairs with a uniform `nn.Module` interface and required one technical contribution during implementation: a selective $\mathrm{SO}(3)$ bispectrum on S^2 achieving the optimal $\Theta(L^2)$ coefficient count (Section 6). Table 2 confirms that the selective forward pass adds negligible overhead to training pipelines ($<$1 ms for all finite-group modules on a single GPU). Across three experimental settings, bispectral pooling provides numerically exact rotation invariance (note that other pooling strategies, such as norm or max pooling, also yield exact invariance; the bispectrum's distinguishing property is its richer, completeness-oriented invariant, not invariance per se) and a data-efficiency advantage over incomplete alternatives in the low-data / moderate-capacity regime, where the $\Theta(L^2)$ invariant map provides a stronger inductive bias that partially compensates for limited training data.

When not to use the bispectrum. The data-efficiency advantage of complete invariants is most pronounced in the low-data, moderate-capacity regime. At high model capacity (Section 5), incomplete alternatives such as max pooling can match or surpass bispectral pooling, likely because the network learns to compensate for missing invariant information. Each bispectrum module is also invariant to a *specific* group (e.g. rotations only, or translations only); for tasks that need invariance to multiple or unknown symmetries, learned invariance via data augmentation or group-equivariant networks with flexible symmetry may be a better fit. Completeness is a property of the *map*, not of downstream task performance: a complete invariant provides a stronger inductive bias, but does not guarantee superior accuracy on every task. Invertibility of the bispectrum for generic signals was first proved by Giannakis [1989]; Kakarala [2012] extended this to compact groups. For $\mathrm{SO}(3)$ on S^2, computational cost grows as $O(L^2)$ per forward pass; at very large band-limits this may become a bottleneck relative to simpler pooling strategies.

Limitations. Inversion for $\mathrm{SO}(3)$ on S^2 is not yet shipped as a library API; only the selective forward map is. The reconstruction experiment of Figure 5 supports completeness of β_{sel} as a working hypothesis, but a closed-form proof of completeness for the selective construction is left to future work; in particular, a full characterisation of the degenerate set and a numerical-conditioning analysis at large band-limits are open. Our experimental comparisons do not include tensor-product architectures (e3nn, MACE) or wavelet scattering, which target different input assumptions and would require non-trivial adaptation to the benchmarks used here.

Future work. Key directions include: implementing $\mathrm{SO}(3)$ inversion from selective bispectral coefficients as a first-class library API; a closed-form completeness proof for β_{sel}; numerical conditioning analysis at large band-limits; broader framework integration (the library is intentionally framework-agnostic and already drops in as an invariant module for architectures built on `escnn` [Cesa et al., 2022] or `e3nn` [Geiger and Smidt, 2022]); and applications to cryo-EM reconstruction and molecular property prediction.

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A Selective $\mathrm{SO}(3)$ Bispectrum – Structural Background

This appendix gives the structural background for the construction of Section 6: the S^2 band-limited setup and notation (Section A.1); the orbit-space dimension lower bound $(L{+}1)^2 - 3$ that motivates the $\Theta(L^2)$ target (Section A.2); the explicit selective index set together with the implementation-grade BFS variant (Algorithm 2) and per-degree counting result (Section A.3); and the parity structure of the scalar bispectrum that dictates the exclusion of certain odd-parity triples in $\mathcal{T}_\ell$ (Section A.4). A short comparison with prior bispectral constructions (Section A.5) and a pixel-space complexity table (Section A.6) close the appendix. We do not claim a closed-form completeness proof in this revision; the empirical reconstruction of Figure 5 is the proof-of-concept evidence the construction is supported by.

Table 7: Asymptotic entry counts in β_{sel}. The constant-size seed block $\mathcal{S}_{\mathrm{seed}}$ collects every triple at degrees $\leq L_{\mathrm{seed}} = \min(L, 4)$; the high-degree selective part allocates exactly $2\ell{+}1$ rows per target degree (the bootstrap block $\mathcal{T}_\ell$ of eq. (8)).

degree range	selected β	total
$\ell \leq 4$	full seed, ≤ 125	$O(1)$
$\ell \geq 5$	$2\ell{+}1$	$O(\ell)$

The total invariant size is therefore $|\mathcal{S}_{\mathrm{seed}}| + \sum_{\ell=5}^{L}(2\ell{+}1) = \Theta(L^2)$, matching the orbit-space lower bound $(L{+}1)^2 - 3$ of Proposition 8 up to a constant factor (Proposition 13).

A.1 Setup and notation

Definition 2 (Band-limited signal on S^2)**.** A signal $f : S^2 \to \mathbb{R}$ is *band-limited* at degree L if its spherical harmonic expansion

$$f(\theta, \varphi) = \sum_{\ell=0}^{L} \sum_{m=-\ell}^{\ell} a_\ell^m \, Y_\ell^m(\theta, \varphi) \tag{5}$$

has $a_\ell^m = 0$ for all $\ell > L$. We write $\mathbf{F}_\ell = (a_\ell^{-\ell}, \ldots, a_\ell^{\ell}) \in \mathbb{C}^{2\ell+1}$ for the coefficient vector at degree ℓ.

For real f, the conjugation symmetry $a_\ell^{-m} = (-1)^m \overline{a_\ell^m}$ holds, so $\mathbf{F}_\ell$ has $2\ell + 1$ real degrees of freedom. The total signal dimension is $\dim = \sum_{\ell=0}^{L}(2\ell + 1) = (L + 1)^2$.

Definition 3 (Rotation action)**.** Under $g \in \mathrm{SO}(3)$, the spherical harmonic coefficients transform as $\mathbf{F}_\ell \mapsto D^\ell(g)\, \mathbf{F}_\ell$, where D^ℓ is the $(2\ell + 1)$-dimensional Wigner D-matrix.

In the irrep notation of Mataigne et al. [2024], $\{\rho_\ell\}_{\ell\geq 0}$ are the (unitary) irreducible representations of $\mathrm{SO}(3)$ with V_{ρ_ℓ} the $(2\ell{+}1)$-dimensional space of degree-ℓ spherical harmonics and $\rho_\ell(g) = D^\ell(g)$; the Fourier coefficient of f at ρ_ℓ is $\mathcal{F}(f)_{\rho_\ell} = \mathbf{F}_\ell$. We use $\mathbf{F}_\ell$ as the working symbol throughout.

Definition 4 (Bispectrum on S^2)**.** For each triple (ℓ_1, ℓ_2, ℓ) satisfying the triangle inequality $|\ell_1 - \ell_2| \leq \ell \leq \ell_1 + \ell_2$, the *bispectral coefficient* is

$$\beta_{\ell_1,\ell_2,\ell} = \sum_{\substack{m_1,\, m_2,\, m \\ m_1+m_2=m}} C^{\ell,\, m}_{\ell_1,\, m_1;\, \ell_2,\, m_2}\, a^{m_1}_{\ell_1}\, a^{m_2}_{\ell_2}\, \overline{a^{m}_{\ell}}, \tag{6}$$

where $C^{\ell,\, m}_{\ell_1,\, m_1;\, \ell_2,\, m_2}$ denotes the Clebsch–Gordan coefficient $\langle \ell_1, m_1;\, \ell_2, m_2 \mid \ell, m\rangle$. Each $\beta_{\ell_1,\ell_2,\ell}$ is a scalar; for real f it is real when $\ell_1{+}\ell_2{+}\ell$ is even and purely imaginary when $\ell_1{+}\ell_2{+}\ell$ is odd (Lemma 14).

Definition 4 is the specialization of the general G-bispectrum (eq. (2)) to the case $G = \mathrm{SO}(3)$ acting on S^2, where the irreps are Wigner-D matrices and the Fourier coefficients are spherical harmonic coefficients a^m_ℓ.

Definition 5 (CG power spectrum, optional diagnostic)**.** For a triple (ℓ_1, ℓ_2, ℓ) satisfying the triangle inequality, write $(\mathbf{F}_{\ell_1} \otimes \mathbf{F}_{\ell_2})\big|_\ell$ for the projection of the Kronecker product $\mathbf{F}_{\ell_1} \otimes \mathbf{F}_{\ell_2}$ onto the D^ℓ-isotypic component, realised by contracting with the CG coefficients $C^{\ell,\, m}_{\ell_1,\, m_1;\, \ell_2,\, m_2}$. The *CG power spectrum entry* is the squared norm of this projection,

$$P_{\ell_1,\ell_2,\ell} := \|(\mathbf{F}_{\ell_1} \otimes \mathbf{F}_{\ell_2})|_\ell\|^2 = \sum_{m=-\ell}^{\ell} \Big| \sum_{\substack{m_1,m_2 \\ m_1+m_2=m}} C^{\ell,\, m}_{\ell_1,\, m_1;\, \ell_2,\, m_2}\, a^{m_1}_{\ell_1}\, a^{m_2}_{\ell_2} \Big|^2. \tag{7}$$

This is a degree-4, real-valued, $\mathrm{SO}(3)$-invariant polynomial in the signal coefficients. The library appends a $O(L^2)$ family of CG power scalars to the selective bispectrum unconditionally for numerical robustness on the real-signal slice (Remark 20); they leave the asymptotic output size unchanged.

Proposition 6 (Size of the full bispectrum)**.** *The number of admissible triples (ℓ_1, ℓ_2, ℓ) with $0 \leq \ell_1, \ell_2, \ell \leq L$ satisfying the triangle inequality $|\ell_1 - \ell_2| \leq \ell \leq \ell_1 + \ell_2$ is $\Theta(L^3)$.*

Proof. For fixed ℓ_1 and ℓ_2, the triangle inequality admits $\ell \in \{|\ell_1 - \ell_2|, \ldots, \min(\ell_1 + \ell_2, L)\}$, giving $\min(\ell_1 + \ell_2, L) - |\ell_1 - \ell_2| + 1$ valid values of ℓ. Summing:

$$N_{\text{full}} = \sum_{\ell_1=0}^{L} \sum_{\ell_2=0}^{L} \big(\min(\ell_1 + \ell_2, L) - |\ell_1 - \ell_2| + 1\big) = \tfrac{1}{3}L^3 + \mathcal{O}(L^2).$$

□

Remark 7 (Invariance). Each $\beta_{\ell_1,\ell_2,\ell}$ is $\mathrm{SO}(3)$-invariant: the Wigner matrices satisfy $D^{\ell_1} \otimes D^{\ell_2} = \bigoplus_\ell D^\ell$ under the CG decomposition, so the triple product contracted with CG coefficients is unchanged by $\mathbf{F}_\ell \mapsto D^\ell(g)\mathbf{F}_\ell$.

A.2 Why $\mathcal{O}(L)$ is impossible

Proposition 8 (Dimensional lower bound)**.** *Any* smooth or polynomial *complete* $\mathrm{SO}(3)$*-invariant for band-limited signals on S^2 at band-limit L must have codomain of dimension at least $(L{+}1)^2{-}3$.*

Proof. Let $\Phi : \mathbb{R}^{(L+1)^2} \to \mathbb{R}^k$ be a smooth (or polynomial) $\mathrm{SO}(3)$-invariant map that separates orbits on a Zariski-open dense subset $U \subset \mathbb{R}^{(L+1)^2}$. Smoothness implies Φ has a well-defined generic rank $r := \max_{x\in U} \operatorname{rank} d\Phi_x$; on the open dense subset where this rank is attained, Φ is locally a submersion onto an r-dimensional submanifold, and orbit-separation forces $r \geq \dim(\mathbb{R}^{(L+1)^2}/\mathrm{SO}(3)) = (L+1)^2 - \dim \mathrm{SO}(3) = (L+1)^2 - 3$ (generic-rank theorem for $\mathrm{SO}(3)$ acting on V_L; cf. Tarski–Seidenberg in the polynomial case). Hence $k \geq r \geq (L{+}1)^2 - 3 = \Omega(L^2)$. □

Remark 9. The chain $\beta_{1,\ell,\ell+1}$ for $\ell = 0, \dots, L-1$ provides only $\sim 3L$ scalar constraints (since each chain entry couples $\mathbf{F}_1 \in \mathbb{C}^3$ with one known $\mathbf{F}_\ell$). This cannot recover $(L+1)^2 \approx L^2$ unknowns.

Remark 10 (Smoothness assumption). The smoothness/polynomial qualifier is essential: pathological non-measurable invariants (e.g. a fixed orbit-section indexed by the axiom of choice) can in principle have codomain dimension 1 but are not implementable. All practical bispectral or spectral invariants are polynomial, so the proposition applies.

A.3 The selective index set

The goal is to select, for each target degree $\ell \geq 8$, exactly $2\ell{+}1$ bispectral triples (ℓ_1, ℓ_2, ℓ) such that the selected entries are simultaneously linear in $\mathbf{F}_\ell$ (or in $\overline{\mathbf{F}_\ell}$, which we conjugate) and have a coefficient matrix depending only on already-recovered lower degrees. Combined with a constant-size full-bispectrum seed at degrees $\leq L_{\text{seed}} = \min(L, 4)$, this gives an $O(L^2)$-sized invariant matching the orbit-space lower bound of Proposition 8.

Definition 11 (Selective SO(3) bispectrum on S^2)**.** Set $L_{\text{seed}} := \min(L, 4)$. The *seed index set* is the full bispectrum truncated at L_{seed},

$$\mathcal{S}_{\text{seed}} := \{(\ell_1, \ell_2, \ell) : 0 \leq \ell_1 \leq \ell_2 \leq L_{\text{seed}},\ |\ell_1{-}\ell_2| \leq \ell \leq \min(\ell_1{+}\ell_2, L_{\text{seed}})\},$$

of size at most $(L_{\text{seed}}{+}1)^3 \leq 125$, independent of L once $L \geq 4$. This matches the implementation seed $L_0 = 4$ of Algorithm 2, on which Bandeira et al.'s exact-arithmetic Gröbner-basis evidence for the scalar bispectrum at low degrees [Bandeira et al., 2017] is strongest. For each intermediate target $5 \leq \ell \leq 7$ we use the small explicit linear block $\mathcal{T}_\ell^{\text{small}}$ tabulated in Algorithm 2; for each high-degree target $\ell \geq 8$ the *bootstrap block* $\mathcal{T}_\ell$ (also used in Proposition 12) is

$$\begin{aligned} \mathcal{T}_\ell := &\{(a, \ell, \ell - a) : 1 \leq a \leq \ell{-}1\} \\ &\cup \big\{(a, \ell, \ell - a + 1) : 2 \leq a \leq \ell{-}1,\ a \neq (\ell+1)/2\big\} \\ &\cup \{(a, \ell - a, \ell) : 1 \leq a \leq 4\} \ \cup\ \mathcal{Z}_\ell, \end{aligned} \tag{8}$$

where $\mathcal{Z}_\ell = \emptyset$ for ℓ even and $\mathcal{Z}_\ell = \{(2, \ell{-}1, \ell)\}$ for ℓ odd ($\ell \geq 9$). The exclusion $a \neq (\ell+1)/2$ for odd ℓ removes the self-conjugate triple $(r, 2r-1, r)$ at $\ell = 2r-1$, which has a repeated index and is identically zero on real signals (Proposition 16); the compensating row $(2, \ell{-}1, \ell) \in \mathcal{Z}_\ell$ is admissible ($|\ell{-}1 - 2| \leq \ell \leq \ell{-}1 + 2$), has all-distinct indices, and is not contained in the chain family above (which only covers $(a, \ell{-}a, \ell)$ with $a = 1, \dots, 4$, i.e. pairs $(a, \ell{-}a)$ with $a \leq 4$). The size counts to $(\ell{-}1) + (\ell{-}2) + 4 = 2\ell{+}1$ for even ℓ and $(\ell{-}1) + (\ell{-}3) + 4 + 1 = 2\ell{+}1$ for odd $\ell \geq 9$. The bispectral index set is

$$\mathcal{S}_\beta := \mathcal{S}_{\text{seed}} \ \cup \bigcup_{\ell=5}^{L} \mathcal{T}_\ell,$$

where $\mathcal{T}_\ell = \mathcal{T}_\ell^{\text{small}}$ for $5 \leq \ell \leq 7$ (Algorithm 2) and is given by eq. (8) for $\ell \geq 8$. The *selective bispectrum* $\beta_{\text{sel}} : V_L \to \mathbb{R}^N$ is the concatenation of all selected scalar entries $\{\beta_{\ell_1,\ell_2,\ell}\}_{(\ell_1,\ell_2,\ell)\in\mathcal{S}_\beta}$; by Lemma 14, each entry contributes one real component on the real-signal slice (real for even parity, purely imaginary for odd parity). The optional CG power augmentation (Definition 5) may be appended at $O(L^2)$ cost without changing the asymptotic budget; the implementation does this unconditionally, see the library implementation note below.

Library implementation note. Algorithm 2 below specifies the implementation-grade index construction used by the `SO3onS2` module. It augments Definition 11 in two ways: (i) it includes *mandatory even self-couplings* $\beta_{\ell,\ell,\ell'}$ (ℓ' even, $2 \leq \ell' \leq \ell$) and the power entries $\beta_{0,\ell,\ell}$ for numerical robustness on the real-signal slice, and (ii) it greedily appends CG-power entries to $\mathcal{S}_P$ to maximise the per-degree real Jacobian rank. The augmented invariant is a strict superset of β_{sel}; this is the structural index set used by the implementation.

Proposition 12 (Bootstrap-block entry count)**.** *For each target degree $\ell \geq 8$, the bootstrap block $\mathcal{T}_\ell$ defined in* (8) *contains exactly $2\ell{+}1$ admissible triples and they are pairwise distinct. Each triple in $\mathcal{T}_\ell$ has ℓ in exactly one of the three slots, with the other two slots filled by indices strictly smaller than ℓ, so each entry contributes one scalar equation linear in either $a_\ell^{m_2}$ or $\overline{a_\ell^m}$ with coefficients depending only on lower-degree $\mathbf{F}_{\ell'}$ ($\ell' < \ell$).*

Proof. The three subfamilies in (8) have sizes $\ell-1$, $\ell-2$, 4, summing to $2\ell+1$. Triangle-inequality and bound checks ($a \le \ell-1$, $\ell-a \ge 1$, $\ell-a+1 \le \ell$) show every triple is admissible. Pairwise distinctness is immediate from the disjoint patterns of the position of the index ℓ. Linearity in the ℓ-slot: the first two families place ℓ in the ℓ_2-slot of (6), so $a_\ell^{m_2}$ enters once with known $a_a^{m_1}$ and $\overline{a_{\ell-a}^m}$ (resp. $\overline{a_{\ell-a+1}^m}$); the third family places ℓ in the ℓ-slot, so $\overline{a_\ell^m}$ enters once with known $a_a^{m_1}$ and $a_{\ell-a}^{m_2}$. □

Proposition 13 (Output size)**.** *For every $L \ge 4$,*

$$|\beta_{\mathrm{sel}}| \;=\; |\mathcal{S}_{\mathrm{seed}}| + \sum_{\ell=5}^{L}(2\ell+1) \;=\; \Theta(L^2),$$

counting one real component per scalar entry (real for even parity, purely imaginary for odd parity, by Lemma 14). This matches the orbit-space lower bound $(L+1)^2 - 3$ of Proposition 8 up to an additive constant (no factor-of-two from complex storage). For $L = 4$ the sum is empty and $|\beta_{\mathrm{sel}}| = |\mathcal{S}_{\mathrm{seed}}| = O(1)$. The count matches the 30 (resp. 49) live entries reported by the implementation at $L = 4$ (resp. $L = 5$).

A.4 Parity structure of the bispectrum

The selective construction of Section 6 exploits two structural facts about the scalar bispectrum $\beta_{\ell_1,\ell_2,\ell}$: a parity rule that fixes whether each entry is real or purely imaginary on real signals (Lemma 14) and the resulting identical vanishing of certain odd-parity entries with a repeated index (Proposition 16). The latter dictates the explicit exclusion in $\mathcal{T}_\ell$ at odd target degrees, and the former is the source of the rank deficit that motivates the CG-power augmentation of Algorithm 2 (Remark 20).

Lemma 14 (Parity of the bispectrum)**.** *For any signal (real or complex):*

$$\beta_{\ell_1,\ell_2,\ell}(T_R \cdot f) = (-1)^{\ell_1+\ell_2+\ell}\,\beta_{\ell_1,\ell_2,\ell}(f). \tag{9}$$

For real *signals, there is an independent constraint:*

$$\overline{\beta_{\ell_1,\ell_2,\ell}(f)} = (-1)^{\ell_1+\ell_2+\ell}\,\beta_{\ell_1,\ell_2,\ell}(f). \tag{10}$$

Together:

(a) If $\ell_1+\ell_2+\ell$ is even: β is real-valued (by (10)*) and T_R-invariant (by* (9)*).*
(b) If $\ell_1+\ell_2+\ell$ is odd: β is purely imaginary (by (10)*) and $\beta(T_R \cdot f) = -\beta(f)$, so $\operatorname{Im}\beta$ flips sign under T_R.*

Proof. Under T_R, each $a_\ell^m \mapsto (-1)^m a_\ell^{-m}$. Substituting into $\beta = \sum_m(\sum_{m_1+m_2=m} C\, a^{m_1} a^{m_2})\overline{a^m}$, the three sign factors $(-1)^{m_1+m_2+m} = (-1)^{2m} = 1$ cancel. Re-indexing $m \mapsto -m$ and applying the CG symmetry $C^{\ell,\,-m}_{\ell_1,\,-m_1;\,\ell_2,\,-m_2} = (-1)^{\ell_1+\ell_2-\ell}\, C^{\ell,\,m}_{\ell_1,\,m_1;\,\ell_2,\,m_2}$ yields (9) (with $(-1)^{\ell_1+\ell_2-\ell} = (-1)^{\ell_1+\ell_2+\ell}$).

For real signals, $\overline{a_\ell^m} = (-1)^m a_\ell^{-m}$. Taking the complex conjugate of β and re-indexing similarly yields (10). □

Lemma 15 (CG power is T_R-invariant)**.** *For all triples (ℓ_1,ℓ_2,ℓ): $P_{\ell_1,\ell_2,\ell}(T_R \cdot f) = P_{\ell_1,\ell_2,\ell}(f)$.*

Proof. Under T_R, the CG-projected component transforms as $(\mathbf{F}_{\ell_1} \otimes \mathbf{F}_{\ell_2})|_\ell^M \mapsto (-1)^{\ell_1+\ell_2+\ell+M}\,(\mathbf{F}_{\ell_1} \otimes \mathbf{F}_{\ell_2})|_\ell^{-M}$. Taking the squared norm and re-indexing the sum over M leaves P unchanged. □

Proposition 16 (Vanishing of odd-parity entries with a repeated index)**.** *Let $\beta_{\ell_1,\ell_2,\ell}$ be the scalar bispectrum entry of a real signal f (so $\ell_1+\ell_2+\ell$ odd). If the index multiset $\{\ell_1,\ell_2,\ell\}$ contains a repeated value, then $\beta_{\ell_1,\ell_2,\ell}(f) \equiv 0$ as a polynomial in the SH coefficients of f.*

Proof. Define the symmetric scalar 3j contraction

$$T_{\ell_1,\ell_2,\ell_3}(f) := \sum_{m_1,m_2,m_3} \begin{pmatrix} \ell_1 & \ell_2 & \ell_3 \\ m_1 & m_2 & m_3 \end{pmatrix} a_{\ell_1}^{m_1}\, a_{\ell_2}^{m_2}\, a_{\ell_3}^{m_3}.$$

Converting the Wigner 3j into a Clebsch-Gordan coefficient via $\left(\begin{smallmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{smallmatrix}\right) = \frac{(-1)^{j_1-j_2-m_3}}{\sqrt{2j_3+1}}\, C^{j_3,\,-m_3}_{j_1,\,m_1;\,j_2,\,m_2}$ and applying the reality identity $a_{\ell_3}^{m_3} = (-1)^{m_3}\,\overline{a_{\ell_3}^{-m_3}}$ shows

$$\beta_{\ell_1,\ell_2,\ell}(f) = (-1)^{\ell_1+\ell_2}\sqrt{2\ell+1}\, T_{\ell_1,\ell_2,\ell}(f),$$

so β and T have the same vanishing locus. The 3j symbol is invariant under even column permutations and picks up $(-1)^{\ell_1+\ell_2+\ell_3}$ under odd ones. If two columns i, j satisfy $\ell_i = \ell_j$, the inserted vectors a_{ℓ_i}, a_{ℓ_j} are literally the same coefficient vector of f, so swapping those two columns leaves the contracted sum T unchanged while multiplying the 3j prefactor by $(-1)^{\ell_1+\ell_2+\ell_3}$. For odd parity this forces $T = -T$, hence $T \equiv 0$ and $\beta \equiv 0$. □

Remark 17 (Consequences for $\mathcal{T}_\ell$). Proposition 16 applies in particular to the family $(r, 2r-1, r)$: it has parity $4r-1$ (odd) and a repeated index, so $\beta_{r,2r-1,r} \equiv 0$ on real signals. Numerical evaluation at the seed-42 real witness confirms this: for $r = 2, \ldots, 7$ the entry is zero to double-precision rounding ($|\beta| \lesssim 10^{-14}$). Any formal definition of $\mathcal{T}_\ell$ must therefore exclude these triples (see (8)); we substitute a chain row with all-distinct indices in their place.

Corollary 18 (First nonzero odd-parity entry)**.** *The first bispectral entry with odd parity and generically nonzero imaginary part occurs at degree $\ell = 4$: $\beta_{2,3,4}$, with $\ell_1 + \ell_2 + \ell = 9$ (odd) and ℓ_1, ℓ_2, ℓ all distinct. At the deterministic witness (seed 42), $\mathrm{Im}(\beta_{2,3,4}) \approx 3.26 \neq 0$, and $\mathrm{Im}(\beta_{2,3,4})(T_R \cdot f) \approx -3.26$. Similarly, $\beta_{2,4,3}$ and $\beta_{3,4,2}$ are nonzero odd-parity entries at degree 4.*

Remark 19 (No parity-breaking at $\ell \leq 3$). For degrees ≤ 3, every odd-parity triple (ℓ_1, ℓ_2, ℓ) has at least two indices equal (since $\max(\ell_1, \ell_2, \ell) \leq 3$ and $\ell_1 + \ell_2 + \ell$ odd forces at least one pair to be equal in this range). By Proposition 16, all odd-parity entries vanish. Therefore, the T_R ambiguity cannot be resolved using only degrees ≤ 3.

Remark 20 (Why the augmented block is needed). The bootstrap block $\mathcal{T}_\ell$ alone is rank-deficient on the real-signal slice: at the seed-42 real witness, the $(2\ell+1) \times (2\ell+1)$ submatrix A_ℓ assembled from the $\mathcal{T}_\ell$ rows has SVD rank $\approx \ell+2$ rather than $2\ell+1$ for $\ell \in \{8, \ldots, 15\}$ (verified numerically; the deficit is generic over real witnesses). The mandatory even self-couplings and CG-power scalars are quadratic or cubic in $\mathbf{F}_\ell$ and supply the additional independent real equations needed to fill out the per-degree real Jacobian; the augmented per-degree block (linear $\mathcal{T}_\ell$ plus quadratic self-couplings plus CG-power scalars) is what Algorithm 2 assembles, and is what the implementation actually solves at degrees $\ell \geq 5$.

A.5 Comparison with existing results

Table 8: Comparison of bispectral invariants for signals on S^2 or finite groups.

Result	Invariant	Size	Complete?
Kakarala [Kakarala, 2012]	Full bispec.	$O(L^3)$	Yes (constr.)
Edidin & Satriano [Edidin and Satriano, 2024]	Full bispec. (band-lim.)	$O(L^3)$	Yes (generic)
Bendory et al. [Bendory et al., 2025]	Freq. marching	$O(L^2)$/shell	Yes (3 shells)
Mataigne et al. [Mataigne et al., 2024]	Sel. G-bispec. (finite)	$O(\lvert G\rvert)$	Yes (exact)
This work	**Sel.** $\mathrm{SO}(3)$ **on** S^2	$\boldsymbol{\Theta(L^2)}$	**Empirical (Section 6.3)**

A.6 Complexity in pixel space

On S^2, a band-limited signal at degree L is sampled on an equiangular grid of size $N_\theta \times N_\varphi \approx 2L \times 2L$, so $P = 4L^2$ and $L = \Theta(\sqrt{P})$.

Output size.

Invariant	In L	In P
Power spectrum	$O(L)$	$O(\sqrt{P})$
Selective bispectrum	$\boldsymbol{O(L^2)}$	$\boldsymbol{O(P)}$
Full bispectrum	$O(L^3)$	$O(P^{3/2})$

The selective bispectrum is *linear* in the number of pixels.

Computation cost.

Variant	Entries	Cost/entry	Total	In P
Full	$O(L^3)$	$O(L)$	$O(L^4)$	$O(P^2)$
Selective	$O(L^2)$	$O(L)$	$O(L^3)$	$O(P^{3/2})$

Concrete example. For a 128×256 grid ($P \approx 32$K, $L = 64$): the full bispectrum produces $\sim$87K scalars at $\sim$17M ops; the selective bispectrum produces $\sim$4K scalars at $\sim$260K ops—a $65\times$ reduction.

B Extended Benchmark Results

This appendix supplements the summary in Table 2 (main text) with scaling plots produced by `benchmarks/benchmark.py` on a single NVIDIA H100 80 GB GPU (PyTorch 2.10).

B.1 Forward-pass timing

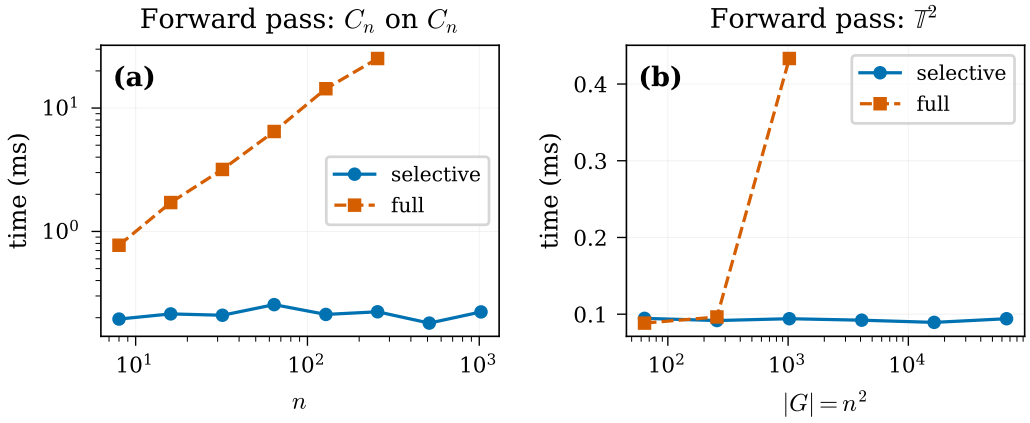

Figure 6: Forward-pass wall-clock time (batch = 16, H100 GPU). **(a)** C_n: the selective pass stays flat at $\sim$0.14 ms regardless of n, while the full bispectrum grows super-linearly (17 ms at n=256). **(b)** $\mathbb{T}^2$: selective is constant at $\sim$0.08 ms up to $|G|$=65,536.

The selective forward pass is dominated by kernel-launch overhead rather than arithmetic, making it effectively free relative to backbone computation. The full bispectrum exhibits the expected super-linear growth and becomes impractical beyond $|G| \approx 256$ for C_n.

B.2 GPU throughput and inversion

All modules exhibit linear throughput scaling with batch size, confirming that the implementation is memory-bandwidth-bound rather than compute-bound at these scales.

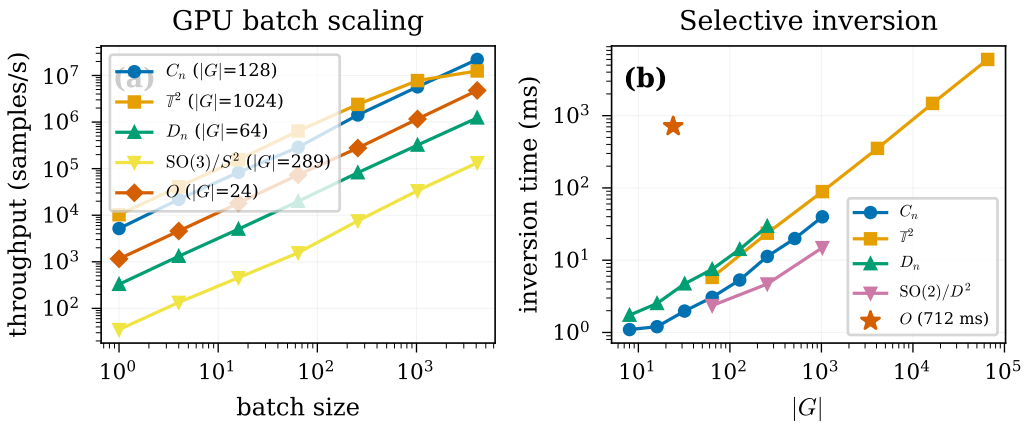

Figure 7: **(a)** GPU throughput (samples/s) vs. batch size for the selective forward pass. C_n and $\mathbb{T}^2$ exceed 10^7 samples/s; the octahedral module reaches 5.7×10^6 samples/s. All modules scale linearly with batch size. **(b)** Inversion wall-clock time. Abelian groups (C_n, D_n, SO(2)/disk) scale roughly linearly. SO(3) on S^2 inversion is not yet implemented.

C PCam: Extended Results

C.1 Published baselines

Table 9 places our results in the context of published methods on PCam and the closely related Camelyon16 benchmark. Our models operate at 68K–920K parameters and are trained on only 10% of the data.

Table 9: Published baselines on PatchCamelyon from Veeling et al. [2018], trained on 100% data.

Method	Reference	Params	ACC	AUC
P4M-DenseNet (C_8)	Veeling et al. 2018	119K	0.898	0.963
P4M-DenseNet M	Veeling et al. 2018	19K	0.893	0.958
DenseNet (augmented)	Veeling et al. 2018	128K	0.881	0.951
DenseNet (no aug.)	Veeling et al. 2018	128K	0.876	0.955

C.2 Model variants

All equivariant models share a C_8-equivariant DenseNet [Huang et al., 2017] backbone with block configuration $(4, 4, 4)$, 24 initial channels, and compression factor 0.5, following the architecture of Veeling et al. [2018] but replacing the final invariant map. Six model variants are evaluated:

- **Standard**: vanilla CNN with random rotation/reflection augmentation and global average pooling. No group dimension, so $\sim 8\times$ fewer parameters at matched growth rate.
- **NormReLU**: C_8-equivariant backbone with norm-based nonlinearity and max pooling over the group dimension.
- **Gated**: C_8-equivariant backbone with gated nonlinearity (doubles internal channels) and group max pool.
- **Fourier-ELU**: C_8-equivariant backbone with Fourier-space ELU nonlinearity and group max pool.
- **Bispectrum**: C_8-equivariant backbone with ReLU nonlinearity and selective G-bispectrum via `CnonCn`(n=8) as the terminal invariant map.

C.3 Training details

To enable a controlled Pareto comparison, we sweep the DenseNet growth rate $k \in \{3, 4, 6, 8, 12\}$ for equivariant models (and $k \in \{6, 12, 20, 30, 35\}$ for the standard CNN to cover a comparable parameter range), yielding models from $\sim$30K to $\sim$1.6M parameters. All models are trained on **10% of training data** (26,214 images) with AdamW [Loshchilov and Hutter, 2019] (lr $= 10^{-3}$, weight decay 10^{-4}), cosine warm restarts, mixed-precision training, and early stopping on validation AUC (patience 10, max 50 epochs).

C.4 Pareto analysis interpretation

Architectural stability (single seed, growth-rate sweep). The bispectrum's AUC varies by only $\sim$1pp across a $12\times$ range of model sizes (80K–920K parameters), while norm swings by $>$7pp. For practitioners, bispectral pooling "just works" without growth rate tuning. The multi-seed matched-parameter results (Table 3) confirm this: bispectrum and Fourier-ELU have the lowest cross-seed variance (± 0.004).

Matched parameters. At matched parameters ($\sim$100K), all equivariant models outperform the standard CNN (AUC 0.896 ± 0.010, 3 seeds). Fourier-ELU leads slightly (0.945 ± 0.004), followed by NormReLU (0.942), bispectrum and gated (both 0.941). All equivariant methods cluster within $\sim$0.5pp of each other.

Scaling behavior. As parameters increase from the smallest to the largest configuration, standard, norm, and Fourier-ELU all degrade (by 1.8, 2.6, and 0.9 AUC points respectively), while the bispectrum improves (+0.9 points, reaching 0.943 at 920K params). This suggests that the bispectrum's complete invariant map scales more gracefully with capacity.

Data scaling (3 seeds). At 1% data the bispectrum leads all methods (0.912 ± 0.025), outperforming NormReLU (0.873), Fourier-ELU (0.876), and gate (0.884). By 10% all equivariant methods converge ($\sim$0.941–0.945). The bispectrum's data-efficiency advantage narrows as data increases, consistent with the hypothesis that complete invariants substitute for missing training examples in the low-data regime.

D OrganMNIST3D: Extended Results

D.1 Published baselines

Table 10 situates our results against published methods on the same benchmark. Our models are deliberately small ($\leq$463K parameters) to enable controlled ablation of the pooling mechanism. The relevant comparison is against equivariant methods at a similar scale.

Table 10: Published baselines on OrganMNIST3D for context. Accuracy differences with respect to large-scale models reflect the deliberate parameter constraint of our ablation study.

Method	Reference	Params	ACC	AUC
ResNet-18 + 3D	Yang et al. 2023	33M	0.907	0.996
ResNet-18 + ACS	Yang et al. 2023	11M	0.900	0.994
ILPOResNet-50	Zhemchuzhnikov and Grudinin [2024]	38K	0.879	0.992
EquiLoPO (local train.)	Zhemchuzhnikov and Grudinin [2025]	418K	0.866	0.991
SE3MovFrNet	Sangalli et al. [2023]	—	0.745	—
Regular SE(3) conv	Kuipers and Bekkers [2023]	172K	0.698	—

D.2 Rotation analysis

Figure 8 shows that all equivariant models achieve identical performance on the original and octahedrally rotated test sets ($\sigma_{\text{rot}} \approx 0$), while the augmentation-trained standard CNN degrades under rotation.

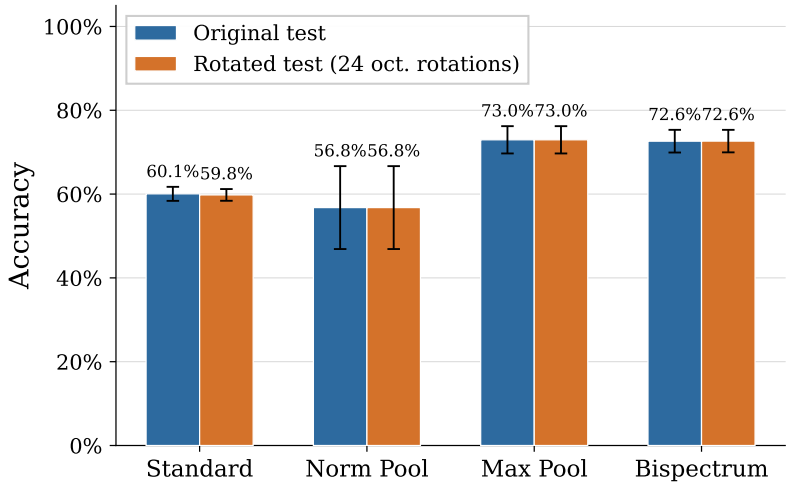

Figure 8: OrganMNIST3D test accuracy on the original and octahedrally rotated test sets. Error bars show standard deviation over three seeds. All equivariant models achieve identical performance under rotation ($\sigma_{\text{rot}} \approx 0$), while the augmentation-trained standard CNN degrades.

D.3 Wider channels

To test whether the advantage of bispectral completeness persists at higher capacity, we run max pool and bispectrum at channel widths $(8, 16)$ and $(16, 32)$ with full training data.

Table 11: OrganMNIST3D accuracy at wider channel configurations (mean $\pm$ std over 3 seeds, full training data).

Channels	Params	Max Pool	Bispectrum
$(4, 8)$	∼375K / 463K	$.730 \pm .033$	$.726 \pm .027$
$(8, 16)$	∼1.5M / 1.7M	$.743 \pm .015$	$.745 \pm .006$
$(16, 32)$	∼6.0M / 6.3M	$\mathbf{.785 \pm .032}$	$.685 \pm .039$

At $(8, 16)$ the two methods are equivalent ($74.5 \pm 0.6\%$ bispectrum vs. $74.3 \pm 1.5\%$ max pool, both at ∼1.5–1.7M parameters). At $(16, 32)$ max pool reaches $78.5 \pm 3.2\%$ while bispectrum drops to $68.5 \pm 3.9\%$—the 6.3M parameter bispectrum model likely overfits the 971 training samples. This suggests that the bispectrum's completeness advantage is most pronounced in the data-limited rather than the parameter-limited regime, consistent with the data efficiency findings in the main text.

E Experimental Details

E.1 Hyperparameters

Table 12 summarises the training hyperparameters for all three experimental settings.

E.2 Hardware

All experiments were run on a single NVIDIA H100 GPU (80 GB). Training wall-clock times range from ∼5 minutes (OrganMNIST3D, 971 samples) to ∼20 minutes (PCam, 26K samples at 10% data). Spherical MNIST bispectrum computation (forward pass through `SO3onS2` at $\ell_{\max} = 15$) takes ∼0.3s per batch of 64 on GPU.

E.3 Data preprocessing

PCam. PatchCamelyon [Veeling et al., 2018] consists of 327,680 RGB patches of size 96×96 pixels extracted from whole-slide images of lymph node sections [Ehteshami Bejnordi et al., 2017].

Table 12: Training hyperparameters across experiments.

	PCam	OrganMNIST3D	Sph. MNIST
Optimizer	AdamW	AdamW	AdamW
Learning rate	10^{-3}	10^{-3}	10^{-3}
Weight decay	10^{-4}	10^{-4}	10^{-4}
Batch size	64–256	16–64	256
Scheduler	Cosine warm restarts	—	—
Max epochs	50	100	50
Early stopping	Val AUC, pat. 10	Val AUC, pat. 15	Val ACC, pat. 10
Seeds	42, 123, 456†	42, 123, 456	42, 123, 456
Mixed precision	Yes	No	No

†Pareto figure uses seed 42; matched-parameter table and data efficiency use seeds 42, 123, 456.

We use the standard train/val/test split. Images are loaded from HDF5 files and normalised to $[0, 1]$. The standard (non-equivariant) baseline uses random rotation and reflection augmentation during training; all equivariant models are trained without augmentation.

OrganMNIST3D. OrganMNIST3D [Yang et al., 2023] contains 1,742 abdominal CT organ volumes of size $28 \times 28 \times 28$ voxels with 11 organ classes (train/val/test = 971/161/610). Volumes are normalised to $[0, 1]$. The standard baseline uses random octahedral augmentation (one of 24 rotations per sample); equivariant models are trained without augmentation. Kernel rotations are implemented as exact voxel permutations on the 3^3 grid, avoiding interpolation artifacts.

Spherical MNIST. MNIST digits are projected onto the sphere via stereographic projection following Cohen et al. [2018] and sampled on a 64×128 equiangular grid. The selective bispectrum is computed at $\ell_{\max} = 15$, producing 384 complex invariant coefficients. The model input is obtained by applying $\mathrm{sign}(x)\log(1 + |x|)$ separately to the real and imaginary parts of each coefficient (a sign-preserving, per-channel injective transform), giving 768 real features. These features are fed to a 3-layer MLP: Linear(768, 256)–BN–ReLU–Linear(256, 128)–BN–ReLU–Linear(128, 10). The power spectrum MLP uses the same structure with hidden=128: Linear(16, 128)–BN–ReLU–Linear(128, 64)–BN–ReLU–Linear(64, 10). The standard CNN has 4 convolutional layers (16–32–64–64 channels) followed by adaptive average pooling and a Linear(1024, 128)–ReLU–Linear(128, 10) classifier.

E.4 Aggregation procedures

PCam. The Pareto figure (Figure 2) uses a single seed (42) across the full growth-rate sweep. The matched-parameter table (Table 3) and data efficiency analysis use 3 seeds (42, 123, 456) per configuration.

OrganMNIST3D. All results in Table 4 report mean $\pm$ std over 3 seeds (42, 123, 456). Rotation robustness (σ_{rot}) is the standard deviation of test accuracy across all 24 octahedral rotations of the test set. The data efficiency analysis (Figure 3 in the main text) uses 3 seeds per data fraction.

Spherical MNIST. All results in Table 5 report mean $\pm$ std over 3 seeds (42, 123, 456). Rotation robustness (σ_{rot}) is measured by applying 10 random $\mathrm{SO}(3)$ rotations to the entire test set and computing the standard deviation of accuracy, yielding 30 measurements per model (10 rotations $\times$ 3 seeds).

E.5 Dataset licenses

- **PCam**: CC0 1.0 Universal (public domain).
- **OrganMNIST3D**: CC BY 4.0.
- **Spherical MNIST**: derived from MNIST, which is publicly available for research use.

F API Example

```python
import torch
from bispectrum import CnonCn, SO2onDisk, OctaonOcta

bsp = CnonCn(n=8, selective=True)
f = torch.randn(16, 8)
beta = bsp(f)
f_rec = bsp.invert(beta)
```

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Algorithm 2 Augmented selective index set for the $\mathrm{SO}(3)$ bispectrum on S^2 at band-limit L. Returns bispectral triples $\mathcal{S}_\beta$ and CG power triples $\mathcal{S}_P$. This extends the BFS-based selective index construction of Mataigne et al. [2024] from finite groups to $\mathrm{SO}(3)$ on S^2, adding the CG power augmentation step that is specific to real signals.

Input: Band-limit L
Output: Bispectral set $\mathcal{S}_\beta$, CG power set $\mathcal{S}_P$
$\mathcal{S}_\beta \leftarrow \{(0,0,0)\}$; $\mathcal{S}_P \leftarrow \emptyset$
Seed block: add to $\mathcal{S}_\beta$ all triples (ℓ_1, ℓ_2, ℓ) with $0 \le \ell_1 \le \ell_2 \le L_0 = 4$ and $|\ell_1 - \ell_2| \le \ell \le \min(\ell_1 + \ell_2, L_0)$ {constant-size full-bispectrum seed}
for $\ell = 1, \ldots, L$ **do**
  budget $\leftarrow 2\ell + 1$
  if $\ell = 4$ **then**
    budget $\leftarrow 10$ {keep all chain+cross at $\ell = 4$}
  end if
  candidates $\leftarrow [\,]$
  if $4 \le \ell \le 7$ **then**
    append the explicit low-degree linear block $\mathcal{T}_\ell^{\mathrm{small}}$ to candidates
  else if $\ell \ge 8$ **then**
    for $a = 1, \ldots, \ell{-}1$ **do**
      append $(a,\ \ell,\ \ell{-}a)$ to candidates
    end for
    for $a = 2, \ldots, \ell{-}1$ **do**
      if ℓ is odd **and** $a = (\ell{+}1)/2$ **then**
        skip $\{(r, 2r{-}1, r) \equiv 0$ on real signals, Prop. 16$\}$
      end if
      append $(a,\ \ell,\ \ell{-}a{+}1)$ to candidates
    end for
    for $a = 1, \ldots, 4$ **do**
      append $(a,\ \ell{-}a,\ \ell)$ to candidates
    end for
    if ℓ is odd **then**
      append $(2,\ \ell{-}1,\ \ell)$ to candidates {compensate excluded $(r, 2r{-}1, r)$ row}
    end if
  else
    append low-degree chain candidates
    if $\ell \ne 2$ **then**
      append cross candidates $\{\beta_{1,2,1}$ excluded$\}$
    end if
  end if
  append $(0,\ \ell,\ \ell)$ to candidates {power entry}
  if $\ell \notin \{1,\ 2\}$ **then**
    for $\ell' = \ell{-}1, \ldots, 0$ **do**
      append $(\ell,\ \ell,\ \ell')$ to candidates
    end for
  end if
  if $\ell = 2$ **then**
    append $(\ell,\ \ell,\ \ell)$ to candidates $\{\beta_{2,2,2}\}$
  end if
  Remove entries that vanish for real signals; deduplicate against $\mathcal{S}_\beta$ already added by the seed block
  $\mathcal{S}_\beta \leftarrow \mathcal{S}_\beta \cup$ first $\min(\text{budget},\ |\text{candidates}|)$ live entries
  {**Mandatory even self-couplings**}
  for $\ell' = 2, 4, \ldots, \lfloor \ell/2 \rfloor \cdot 2$ with $\ell' \le \ell$ and $\ell \ge 3$ **do**
    $\mathcal{S}_\beta \leftarrow \mathcal{S}_\beta \cup \{(\ell,\ \ell,\ \ell')\}$
  end for
  {**CG power augmentation**}
  Greedily add $P_{\ell_1,\ell_2,\ell'}$ (with $\max(\ell_1, \ell_2) = \ell$, $|\ell_1{-}\ell_2| \le \ell' \le \ell_1{+}\ell_2$) to $\mathcal{S}_P$ until the Jacobian rank of Φ_{aug} at degree ℓ matches the number of unknowns
end for
return $(\mathcal{S}_\beta,\ \mathcal{S}_P)$